INCH-POUND

A-A-52093
March 14, 1991

COMMERCIAL ITEM DESCRIPTION

DETERGENT, DRY CLEANING SOLVENT

The General Services Administration has authorized the use of this commercial item description in preference to MIL-D-12150.

Abstract. This commercial item description covers one type and grade of detergent for use with solvents in the dry cleaning of fabrics and clothing.

1. SALIENT CHARACTERISTICS

1.1 Materials. The dry cleaning detergent shall consist of a minimum of 94 percent active detergent plus a liquid nonionic surface active agent. The detergent shall be free of foreign matter and shall not phase separate.

1.1.1 Base material. The dry cleaning detergent shall be 90 percent biodegradable and shall not have a rancid or acrid odor. Residual odor shall be determined in accordance with ASTM D 2553.

1.1.2 Appearance. The detergent shall be homogeneous and free of undissolved matter. A thoroughly mixed sample shall not show phase separation after standing for 24 hours at room temperature.

1.1.3 Flash point. The detergent shall have a minimum flash point of 175°F when tested as specified in ASTM D 93.

1.1.4 Moisture content. The moisture content shall not exceed 4.0 percent when tested as specified in ASTM D 460.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-IRT, Natick, MA 01760-5019.

AMSC N/A

FSC 7930

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1.1.5 Solubility and pH value. The detergent shall be 100 percent soluble and shall not show any separation after 2 hours when tested as follows: Dilute a 4 milliliter sample of dry cleaning detergent in 96 milliliters of type I solvent of P-D-680, and a 4 milliliter sample in tetrachloroethylene of O-T-236. Any evidence of separation is unacceptable. The pH value shall be between 6.0 and 9.0 when tested as specified in ASTM E 70 for 10 percent emulsion.

1.1.6 Soil suspension value. The nonionic detergent shall have a minimum soil suspension value of 1.0 when tested as follows:

Condition eight 4 by 4 inch swatches of white wool fabric (see 4.3) for 48 hours in a desiccator over a saturated solution of $Ca(NO3)$ at room temperature. Place 400 milliliters of type I dry cleaning solvent as specified in P-D-680 containing 0.5 percent by volume of the dry cleaning detergent under test in each of two stainless steel cannisters (3 by 5 inches). Into each of two other cannisters, introduce 400 milliliters of type I dry cleaning solvent as specified in P-D-680 containing no dry cleaning detergent. Into each of the four jars add 0.200 grams of L. Martin's Globe Brand Carbon Black or equal (see 4.4), and disperse for 3 minutes with a high speed stirrer. Place all four jars in a Launder-Ometer at room temperature and operate for 10 minutes. Introduce two white conditioned swatches into each cannister and operate the Launder-Ometer for an additional 20 minutes at room temperature. Remove the swatches from the cannisters, place upon a screen to drain, and air dry on paper towels. Measure the light reflectance on each side of the swatches with a Hunter reflectometer or equivalent instrument, and average the readings.

$$\text{Soil suspension value} = \frac{\text{Average reflectance of fabric using detergent}}{\text{Average reflectance of fabric using no detergent}}$$

1.2 Label. Unless otherwise specified in the contract or purchase order, each shipping container shall be labeled in accordance with the Federal Hazardous Substances Labeling Act. The labeling shall read as follows:

CAUTION - COMBUSTIBLE LIQUID

THIS PRODUCT IS COMBUSTIBLE AND MUST NOT BE USED OR STORED IN AREAS WHERE THERE ARE OPEN FLAMES, ARCING EQUIPMENT, HOT SURFACES, OR WHERE SMOKING IS PERMITTED. AVOID INHALING VAPORS. DO NOT GET IN EYES OR ON SKIN. IN CASE OF CONTACT WITH EYES OR SKIN, IMMEDIATELY FLUSH WITH PLENTY OF WATER FOR AT LEAST 15 MINUTES. FOR EYES, GET PROMPT MEDICAL ATTENTION.

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2. QUALITY ASSURANCE

2.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3. PACKAGING

3.1 Packaging. One gallon of detergent shall be packaged in a plastic container having a reclosable closure. The closure shall be fitted with a heat sealed inner seal or shall be externally secured in a manner to prevent leakage.

3.2 Packing. Six containers of detergent, packaged as specified, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The shipping container shall be fitted with full top and bottom pads and half slotted interlocking partitions fabricated from grade 200 fiberboard. Shipping containers shall be closed in accordance with the appendix of PPP-B-636.

3.3 Palletization. When specified (see 4.1), detergent, packaged as specified in 4.1, shall be palletized in accordance with load type Ia of MIL-STD-147. Pallet type shall be I (4-way entry), type IV, or type V of MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G.

3.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

4. NOTES

4.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. When palletization is required (see 3.3).

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4.2 Sources of documents.

4.2.1 Source of nongovernment documents.

ASTM D 93	-	Test Methods for Flash Point by Pensky-Martens Closed Tester
ASTM D 460	-	Sampling and Chemical Analysis of Soaps and Soap Products
ASTM D 2553	-	Test for Detecting Odor of Drycleaning Detergent on a Drycleaned Garment
ASTM E 70	-	Test Method of pH of Aqueous Solutions With the Glass Electrode

All are available from:

American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103-1187

4.2.2 Source of government documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

4.3 Wool cloth. The white wool fabric swatches for testing should be 100 percent virgin, unbleached wool with approximately 47 ends and 41 picks per inch, and 6 ounces per square yard (203 g/m^2).

4.4 Carbon black. Globe Brand Carbon Black, supplied by the L. Martin Company, Inc., 41 East 42nd Street, New York, New York 10017, has been found to be satisfactory in performing the soil suspension value testing.

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MILITARY INTERESTS:

Custodians

Army - GL
Air Force - 99

Review activities

Army - MD
Air Force - 84
DLA - GS

User activities

Army - AR
Navy - MC, SH

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

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